Homogenouse Transformations

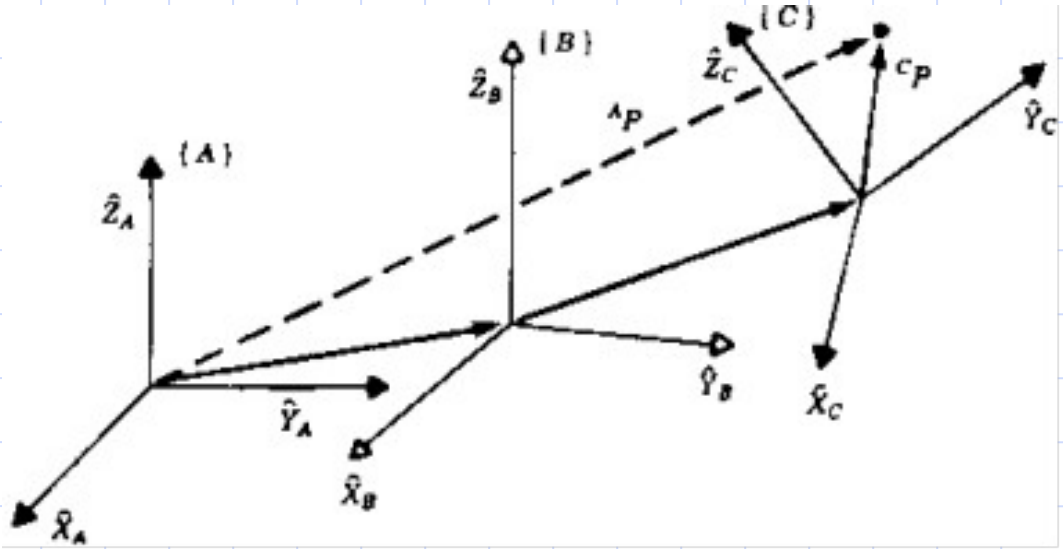
$$^{B}P = {}^{B}_{C}T . {}^{C}P$$

$$^{A}P = {}^{A}_{B}T . {}^{B}P$$

then

$$^{A}_{C}T = {}^{A}_{B}T . {}^{B}_{C}T`$$

$$^{A}P = {}^{A}_{B}T . {}^{B}_{C}T . {}^{C}P$$



$$ {}^{A}_{C}T = \begin{bmatrix} {}^{A}_{B}R\,{}^{B}_{C}R & {}^{A}_{B}R.{}^{B}P_{Corg} + {}^{A}P_{Borg} \\ 0 \quad 0 \quad 0 & 1 \end{bmatrix} $$

$$ {}^{B}_{A}R = {}^{A}_{B}R^{T} $$

$$ {}^{B}\left({}^{A}P_{Borg}\right) = {}^{B}_{A}R.{}^{A}P_{Borg} + {}^{B}P_{Aorg} $$

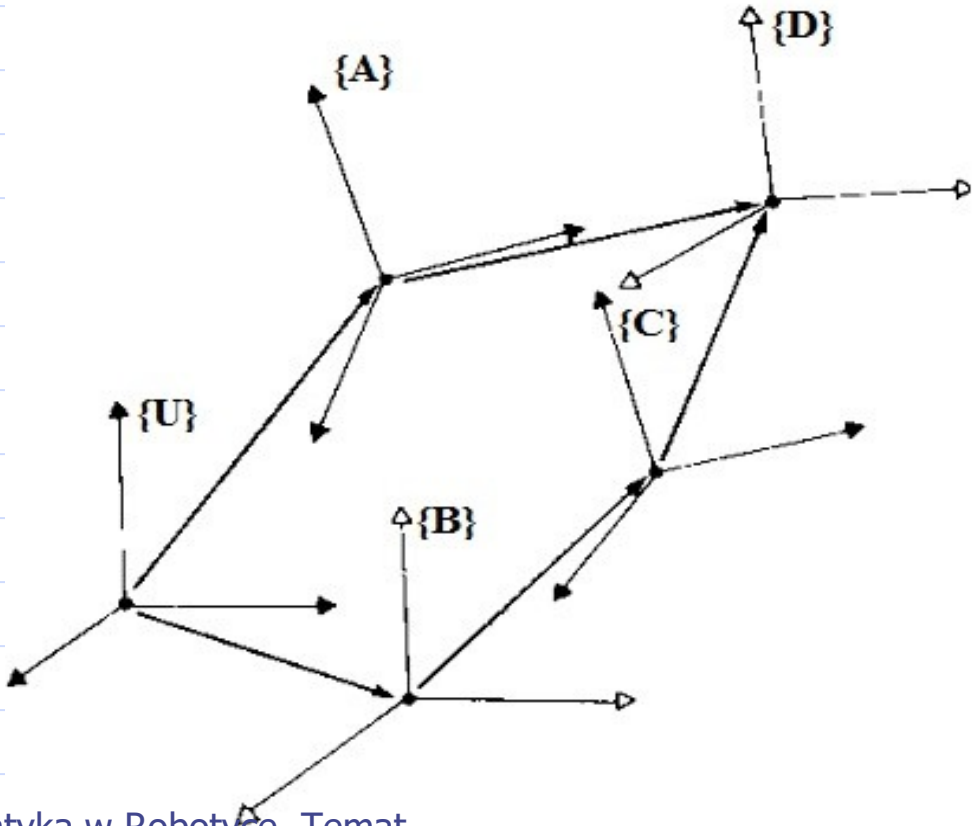
$$ {}^{B}_{A}T = \begin{bmatrix} {}^{A}_{B}R^{T} & -{}^{A}_{B}R^{T}.{}^{A}P_{Borg} \\ 0 \quad 0 \quad 0 & 1 \end{bmatrix} $$

We have: ${}^{U}_{D}T = {}^{U}_{A}T\,{}^{A}_{D}T$, then: ${}^{U}_{D}T = {}^{U}_{B}T\,{}^{B}_{C}T\,{}^{C}_{D}T$

$${}^{U}_{A}T\,{}^{A}_{D}T = {}^{U}_{B}T\,{}^{B}_{C}T\,{}^{C}_{D}T$$

$${}^{A}_{D}T = {}^{U}_{A}T^{-1}\,{}^{U}_{B}T\,{}^{B}_{C}T\,{}^{C}_{D}T$$

$${}^{B}_{C}T = {}^{U}_{B}T^{-1}\,{}^{U}_{A}T\,{}^{A}_{D}T\,{}^{C}_{D}T^{-1}$$

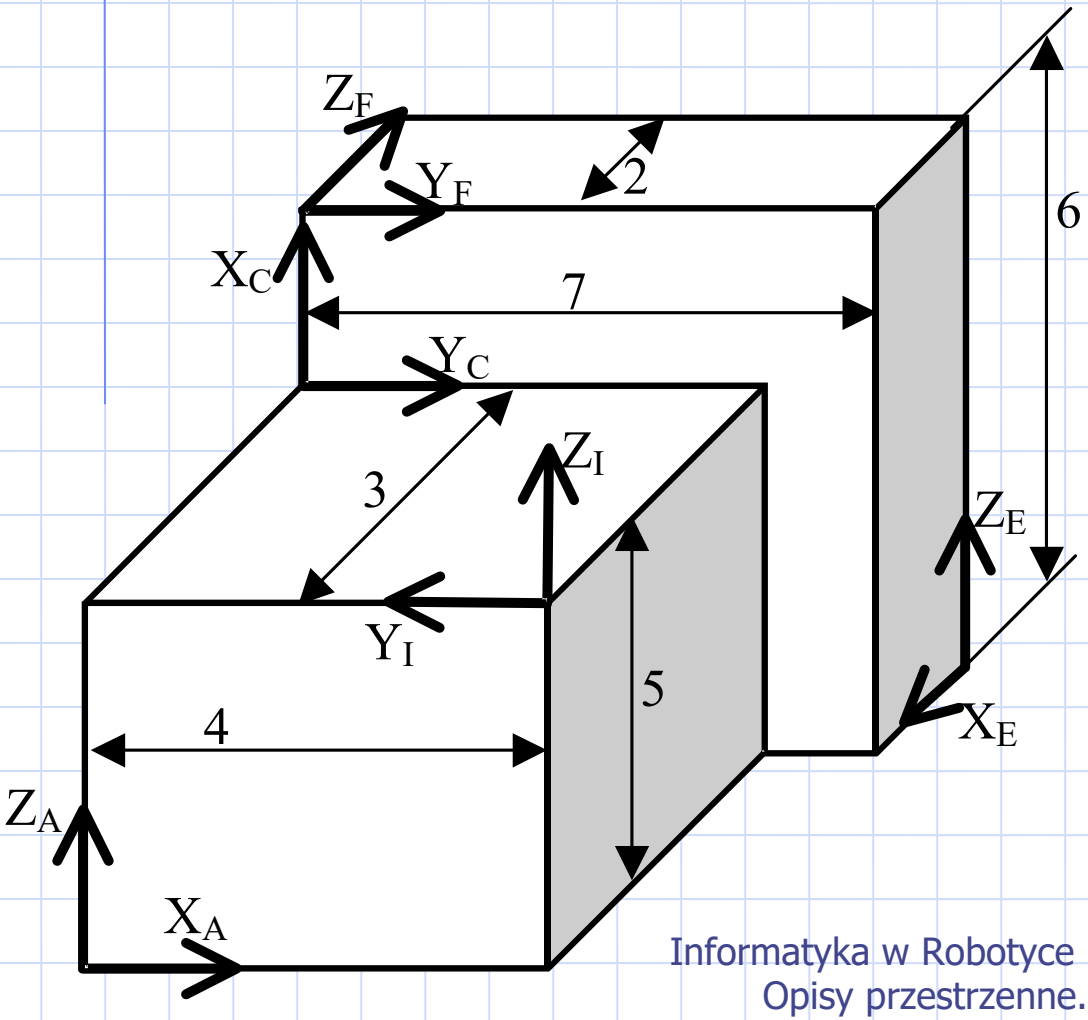


For the next figures:

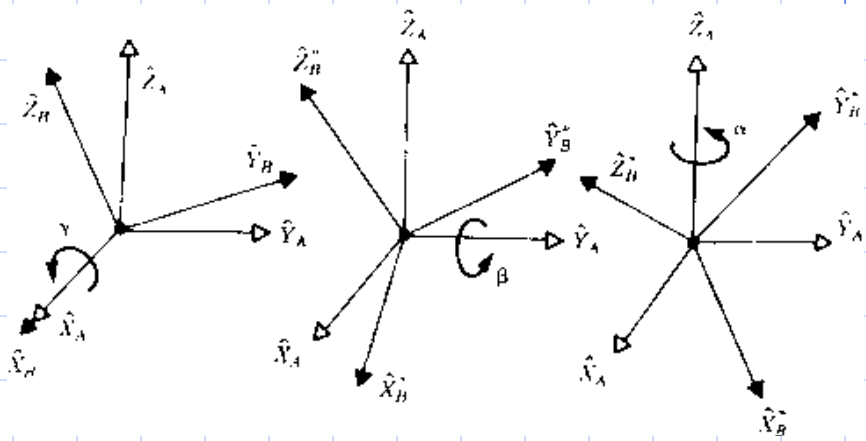
- Draw the missing axis
- Calculate:

$${}^{A}_{E}T \quad {}^{C}_{I}T \quad {}^{F}_{A}T$$

$${}^{A}_{B}T \quad {}^{B}_{A}T \quad {}^{D}_{B}T \quad {}^{A}_{E}T$$



Angles X-Y-Z



$$ {}^{A}_{B}R_{XYZ}(\gamma,\beta,\alpha) = R_Z(\alpha)R_Y(\beta)R_X(\gamma) = $$

$$ = \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix} $$

$$ {}^{A}_{B}R_{XYZ}(\gamma,\beta,\alpha) == \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} $$

Rotation Angles on X-Y-Z cont.

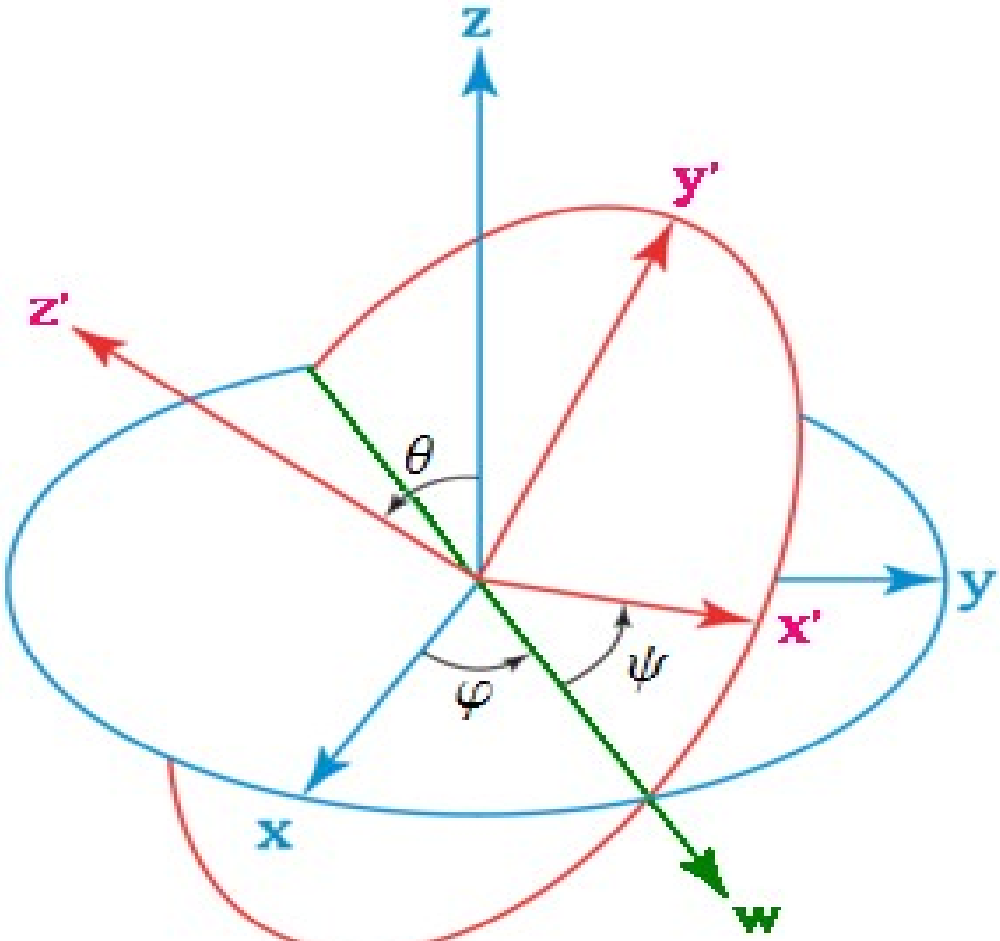
$$ {}^{A}_{B}R_{XYZ}(\gamma,\beta,\alpha) = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

$$ \beta = A\tan 2\left(-r_{31}\sqrt{r_{11}^2 + r_{21}^2}\right) $$

$$ \alpha = A\tan 2\left(r_{21}/c\beta,\quad r_{11}/c\beta\right) $$

$$ \beta = A\tan 2\left(r_{32}/c\beta,\quad r_{33}/c\beta\right) $$

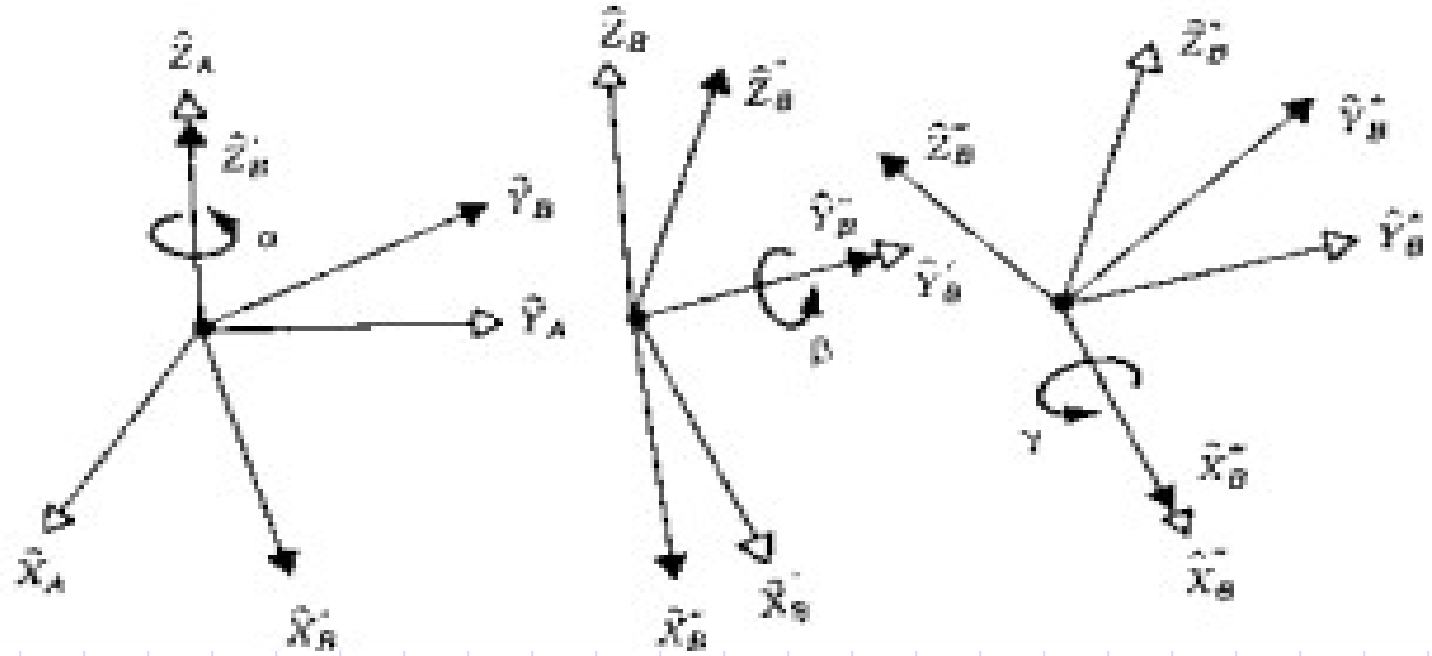
Kąty Eulera dla prawoskrętnego układu współrzędnych



- φ — kąt mierzony od osi **x** do osi węzłów **w** w kierunku wyznaczonym osią **z**; jest to kąt obrotu 1.
- θ — kąt mierzony od osi węzłów **w** do osi **x'** w kierunku wyznaczonym osią **z'**; jest to kąt obrotu 3.
- ψ — kąt mierzony od osi **z** do **z'** w kierunku wyznaczonym osią węzłów **w**; jest to kąt obrotu 2.

Eulera Angles (Roll-Pitch-Yaw)

Z-Y-X (PRY)



$$ {}^{A}_{B}R_{Z'Y'X'} = R_Z(\alpha)R_Y(\beta)R_X(\gamma) = $$

$$ = \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix} $$

$$ {}^{A}_{B}R_{Z'Y'X'}(\alpha,\beta,\gamma) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} $$

$$R_k(\theta) = \begin{bmatrix} k_x k_x v\theta + c\theta & k_x k_y v\theta - k_z s\theta & k_x k_z v\theta + k_y s\theta \\ k_x k_y v\theta + k_z s\theta & k_y k_y v\theta + c\theta & k_y k_z v\theta - k_x s\theta \\ k_x k_z v\theta - k_y s\theta & k_y k_z v\theta + k_x s\theta & k_z k_z v\theta + c\theta \end{bmatrix}$$

